PRIMES-Conjectures: State of the Project and Findings

Two-page summary, 2026-08-27. Full record: README.md (literature), PLAN.md (plan v2), data/FINDINGS.md (all measurements), doc/registry.md (component catalog H1-H8 / S1-S7 / V1-V5).

1. Goal and background

The project attacks **Agrawal's conjecture** (from the AKS 'PRIMES is in P' line of work): if r is a prime not dividing n and $(X-1)^n = X^n-1$ (mod n, X^r-1), then n is prime or $n^2 = 1$ (mod r). If true, it yields a deterministic O~($\log^3$ n) primality test, beating the unconditional state of the art (Lenstra-Pomerance 2019, O~($\log^6$ n)). The conjecture is open but believed false: the Lenstra-Pomerance (L-P) heuristic constructs counterexamples as squarefree $n = p_1 \ldots p_k$ whose prime factors satisfy p = 3 (mod 80), p-1 | n-1, and p+1 | n+1 - making n simultaneously a Carmichael and Lucas-Carmichael number. No such number is known below 10^{18}, and exhaustive search (Primaboinca BOINC project) is clean below 10^{17}; heuristics say counterexamples exist but are hundreds of digits long. The project's aim was to **build one constructively** instead of searching, using twin-smooth integers. Popovych's 2008 strengthened conjecture (adding an $(X+2)^n$ condition) is tracked as a side question.

2. The pipeline: harvest, then solve

The working paper's reduction (Prop. 3.1, machinery unit-tested in Phase 0) turns the L-P construction into a two-stage pipeline. **Harvest:** collect pool primes p = 2m+1 where m = 1 (mod 40) and (m, m+1) is a twin-smooth pair over disjoint prime sets $Q_- \mid Q_+$, so that p-1 and p+1 are automatically smooth over the two sides. **Solve:** find an odd-size subset S of the pool whose product is = 1 (mod Lambda_-) and = -1 (mod Lambda_+), where $\text{Lambda}_{+/-}$ = lcm of the p-/+1 values; n = product of S is then a counterexample. This is a 0/1 subset-product problem in the abelian group G = $(Z/\text{Lambda}_-)^*$ x $(Z/\text{Lambda}_+)^*$. Three go/no-go gates were defined: G1 (supply exists), G2 (supply exceeds demand with margin), G3 (solver works on synthetic instances of real dimensions).

3. What succeeded: the supply side is solved

- **Twin-smooth supply curves anchored.** A compiled CHM-closure harvester (code/chm_closure.c) reproduces the published complete twin-smooth datasets to 99.2-99.7% with exact largest elements (B=100: 13,333/13,374; B=200: 346,110 in ~3 min). Growth fitted at N(B) ~ B^5, much faster than the initial guess. The full 82,026,426-pair B=547 corpus (shared privately by M. Naehrig) was streamed and audited.
- **Dataset-mining is dead - the 0.06 invariant.** Across seven measurements (B = 100 to 547, pools of 78 up to 73,163 elements, two independent optimizers), the best side-partition of organically harvested twins keeps only ~0.06 elements per demand-bit; feasibility needs ~1-3. Measured, not extrapolated, on the entire corpus.
- **Partition-first harvesting bypasses the invariant (G1, G2 passed).** Fixing the partition first and enumerating candidates m over Q_- only makes every hit compatible by construction. Measured supply/demand ratios of 13.8-44x (vs. the 2.5 gate), model validated conservative (0.68x) against the complete B=547 corpus. A production spec was frozen (doc/harvest-spec.md): ratio 13.8, pool ~2.5×10^5, demand ~18,000 bits, a few days on 8-16 cores.
- **Harvest/solver co-design works.** Constraining both factor bases to odd-smooth-shifted primes shrinks the solver group's odd-prime support from 336 primes up to 6269 down to 24 primes up to 97, while keeping the harvest ratio at 7.7 - still above the gate.

4. What failed: the solver wall, quantified

G3 was never passed; the production harvest (A3) was deliberately held until it did, and the investigation instead produced a chain of increasingly sharp negative results:

- **A4:** the group's 2-part is GF(2)-linear and solves at full scale (4,586 components in 12 s) - but is only ~16% of the demand bits. The ~84% odd-prime part (~15,000 bits) is a genuinely hard subset-sum.
- **A6:** generic solvers (annealing, exact meet-in-the-middle) cap 2-3 orders of magnitude below the required dimension.
- **A7/A8:** the right tool was identified - the AGHS subexponential omega-guided subset-product method (used to build a Carmichael number with 10^{10} prime factors). But it requires pool elements that are the identity on *many* group components (identity-density rho ~ 0.3-0.5).
- **N1:** the harvest paradigm that gives AGHS-like density (divisor paradigm: p-1 | Lambda_- and p+1 | Lambda_+ for fixed moduli) is **unharvestable** under this double condition - pools of 0-2 elements at every feasible scale, because it is the product of two rare events (AGHS needs only the single condition p-1 | Lambda).
- **A10:** a correct Wagner-4 generalized-birthday solver was built and its reach measured as a function of density: r_{max} ~ log(N)/(1-rho). Reaching our ~1000-component groups needs rho >= 0.98 (optimistically 0.85-0.95 for higher-order Wagner); harvestable pools give rho ~ 0.004, and growing the pool helps only logarithmically. The 'maybe the solver tolerates low density' escape is **closed with a measured threshold**.

- **A11:** no other AKS modulus r helps. Every r = 1 (mod 4) gives exactly the r=5 structure (the same double condition on p^2-1), and every other r is strictly worse (larger cyclotomic factors to make smooth). A single-condition (degree-1) variant is impossible for any valid r. The obstruction is intrinsic to the AKS construction, not a quirk of r=5.

Route	Status	Where it is settled
Dataset mining (harvest twins, then partition)	Dead: 0.06 invariant, ~16-25x below feasibility	FINDINGS: B=547 full-corpus measurement
Partition-first harvest (supply)	Solved: ratio 13.8-44x, spec frozen	A1/A2, gates G1+G2 passed
High-density harvest for AGHS solver (H8)	Blocked: unharvestable under double condition	N1
AGHS-class solver on harvestable pools	Blocked: needs rho >= 0.98, pools give 0.004	A10 (measured reach law)
Friendlier modulus r (option c)	Blocked: double condition intrinsic to every r	A11
New low-density solver paradigm (option b)	Open (long shot); problem brief written	doc/option-b-problem.md

5. Bottom line and current state

The twin-smooth / Lenstra-Pomerance route to an Agrawal counterexample is blocked at exactly one step, by a **two-part quantified obstruction**: pools dense enough for the only scalable solver family cannot be harvested (N1), and pools that can be harvested cannot be solved (A10) - and this holds for every AKS modulus r (A11). This is not a disproof of the conjecture, and not a claim that no counterexample can ever be built; it is a measured infeasibility result for this pipeline, which was anticipated from the start as a publishable outcome in its own right.

Work queued or in flight:

- **Negative-result paper (option d, main deliverable).** A complete execution plan for an 8-12 page arXiv-style technical note exists (doc/negative-result-plan.md), scoped for a separate agent/branch; every number traces to dated FINDINGS sections.
- **Option (b) research brief.** The one live (long-shot) direction: a solver for 0/1 subset-product in ~24,000-bit smooth-exponent groups with dense vectors and only ~linear pool/dimension ratio, posed as an open problem with a regression harness (doc/option-b-problem.md).
- **Theory track.** B1 (independent re-derivation of the L-P theorem, which everything leans on) remains open; B2 (other admissible local cells) and B4 (what Popovych's extra condition demands of any find) are unstarted. Outreach drafts to Costello/Naehrig are in doc/.

Assets worth keeping regardless of route: the validated C CHM-closure harvester, the near-complete twin-smooth corpora and supply-curve fits, the frozen partition-first harvest spec, the GF(2) 2-part solver, the density-reach measurement methodology, and the component registry that makes all of it recombinable.